

Analysis of 3rd-party Data Sharing and Data Tracking and of GDPR Compliance of Norwegian Web Sites

Humna Akhtar, Mithun Chandra Debnath, and Karina Sætersdal Nilssen

Department of Computer Science, Oslo Metropolitan University

ACIT4280 Privacy by Design

Group Assignment 1A

3 September 2026

Introduction

This report answers two questions. When a person opens a Norwegian-facing front page, how many other companies does that page contact, and in which countries do those machines appear to sit? When a site uses personal data, has it named a GDPR Article 6 basis,¹ and does that basis hold in the ICO tool? The questions differ. A news site can load many ad scripts and still have a contract for a paid account. A tax office can lawfully hold income data and still fail a consent test for optional statistics cookies. Part 1 measures contact. Part 2 tests one purpose at a time.

Table 1 lists the 18 course sites. Webbkoll scanned all 18.² Table 1a lists five for the ICO tool.³ Counts: Tables 2 and 3. Sectors: Figures 2 to 4. ICO: Table 5.

Key terms

Personal data identifies a person (name, email, online identifier) (General Data Protection Regulation, 2016). The GDPR is the EU/EEA rule for that information and applies in Norway. *Processing* is collecting, storing, showing or sending it. It is lawful only if an Article 6 basis fits the purpose (General Data Protection Regulation, 2016, art. 6): legal obligation, public task, contract, consent, or legitimate interests.

A *cookie* is a small file on the device. Internal (first-party) cookies come from the site opened. External (third-party) cookies come from someone else, such as an advertiser.

A *third-party request* is a call from the page to another host (fonts, ads, video).⁵ It is not a third-party cookie. A page can call many hosts and set zero such cookies. We rank the 18 sites on requests (Table 2).

Table 1

Course Web Site List Analysed with Webbkoll

Sector (as on Canvas)	Services
Shopping	xxl.no, cdon.no, ikea.no, babyshop.no
Government	skatteetaten.no, barnevernet.no, altinn.no, lanekassen.no
News & media & entertainment	aftenposten.no, nrk.no, netflix.no, document.no, klassekampen.no, spotify.no, worldofwarcraft.com
Spare time / sports	dnt.no, fotball.no, skiforeningen.no

Note. Sectors and service names follow the course assignment list. Where Webbkoll followed a redirect, the original URL is retained (for example, ikea.no to ikea.com).

Table 1a

Five Services Chosen from Table 1 for the ICO Lawful Basis Tool

Service	Sector	Why this site
babyshop.no	Shopping	Retail checkout; high request count on the repeat scan
skatteetaten.no	Government	Public authority; tax and cookies as two purposes
netflix.no	News & media & entertainment	Paid subscription (contract)
document.no	News & media & entertainment	Widest country list; advertising analytics
fotball.no	Spare time / sports	Highest third-party request count in Table 2

Note. Five services selected for ICO analysis, with two from news and media. Full results appear in Table 5.

Part 1: Measuring third-party contact on page load

Webbkoll: tool and scope

Webbkoll is a free scanner (5th of July Foundation, n.d.-c). It opens the URL in a clean Chromium session⁶ and reports HTTPS,⁷ cookies, third-party requests and the server IP⁸ (5th of July Foundation, n.d.-a, n.d.-b). Figure 1 is the start page. It does not read the privacy notice or decide Article 6. It records what one load contacted.

Scan methodology

English Webbkoll on every Table 1 name. One live visit: no add-ons, no Do Not Track.⁹ Two visits on 20 August 2026. Table 2 is the first visit and the ranking. Table 3 is the later visit. We do not average them.

KeyCDN: country lookup and country cells

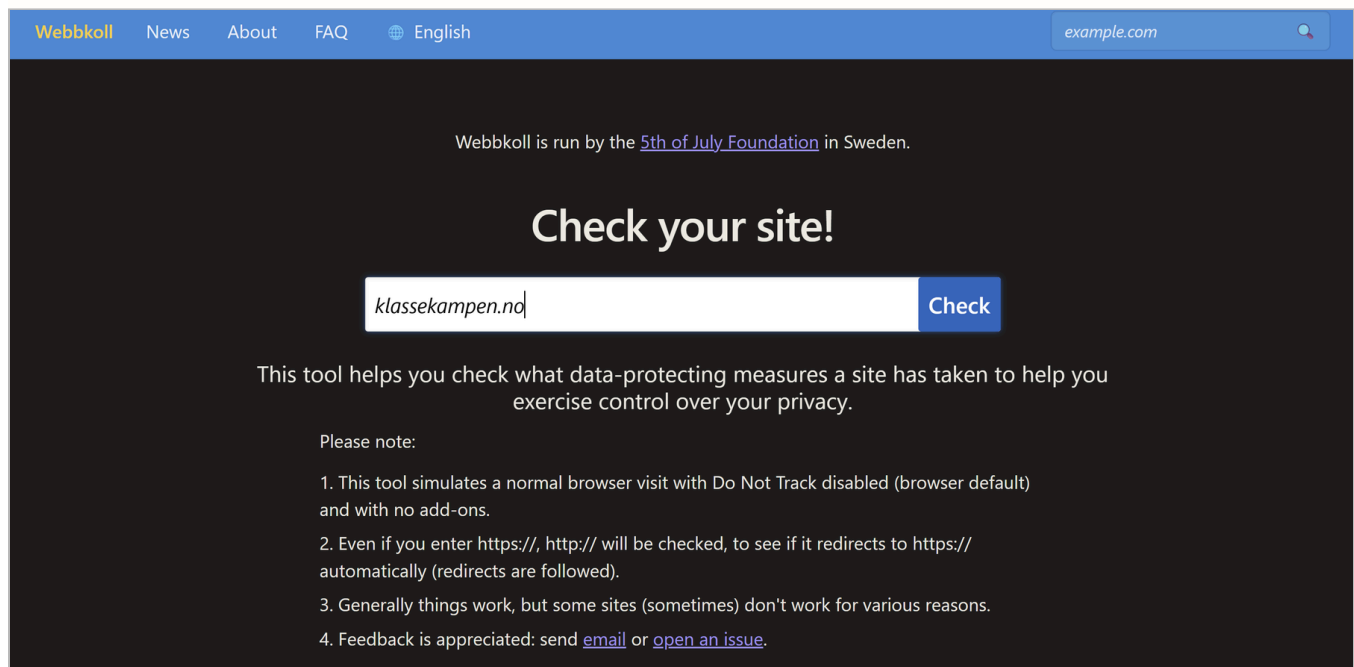
KeyCDN IP Location Finder returns a likely country and provider for an IP address (KeyCDN, n.d.).¹⁰ Location of server uses the site IP. List of countries uses the third-party IPs Webbkoll listed. We count unique countries after dropping Norway. n/a means no country returned.

Webbkoll output columns

Table 2 is one homepage load. Internal cookies: set by the site. External cookies: set by others. Third-party requests: calls to other hosts. Countries excl. Norway: unique countries of those hosts after dropping NO. The list may still print NO; the count already excludes it.

Figure 1

Webbkoll Start Page for *klassekampen.no*

The screenshot shows the Webbkoll website interface. At the top, there is a blue navigation bar with links for 'Webbkoll', 'News', 'About', 'FAQ', and 'English'. A search bar on the right contains 'example.com'. The main content area has a dark background. It starts with the text 'Webbkoll is run by the 5th of July Foundation in Sweden.' followed by a large heading 'Check your site!'. Below this is a search input field containing 'klassekampen.no' and a blue 'Check' button. A paragraph explains the tool's purpose: 'This tool helps you check what data-protecting measures a site has taken to help you exercise control over your privacy.' Below this, a 'Please note:' section lists four points: 1. The tool simulates a normal browser visit with Do Not Track disabled and no add-ons. 2. Both https:// and http:// will be checked for redirects. 3. Things generally work, but some sites may not work for various reasons. 4. Feedback is appreciated, with links to 'email' and 'open an issue'.

Note. Screenshot from *Webbkoll* (5th of July Foundation, n.d.-c; <https://webbkoll.dataskydd.net/en>). *klassekampen.no* is used as the method example because it ranks fifth on third-party requests in Table 2.

Tables 2 and 3: first and repeat measurements

Table 2 is the first scan, 20 August 2026. Read it with Figures 3 to 5. Table 3 is the later visit that day, with third-party countries on 21 August 2026. Table 3 is a check, not a second ranking.

Figures 2 to 4: request counts and sector totals

Figure 2 ranks request counts. Figure 3 is the same counts as a pie. Figure 4 groups them by sector: news/media highest, government lowest, sport high because of fotball.no, shopping in between. Seventeen sites set zero external cookies. ikea.no set two after redirect to ikea.com. Table 2 requests run from 1 (lanekassen.no) to 113 (fotball.no). babyshop.no is 101 on Table 3; rankings stay on Table 2. A high request count is not a cookie count.

Table 2*Webbkoll Results for the 18 Services in Table 1 (First Measurement, 20 August 2026)*

Name of service	Sector	Location of server	Internal cookies	External cookies	3rd-party requests	Countries excl. Norway	List of countries
xxl.no	Shopping	US	3	0	26	4	PL, IE, CA, SE
cdon.no	Shopping	CA	4	0	33	3	CA, US, SE
ikea.no	Shopping	CA	8	2	15	2	CA, US
babyshop.no	Shopping	CA	1	0	45	2	SE, US
skatteetaten.no	Government	CA	5	0	14	5	IE, SE, FR, GB, NO, CA
barnevernet.no	Government	NO	2	0	25	4	SE, DE, CA, NO, GB
altinn.no	Government	CA	0	0	5	1	US
lanekassen.no	Government	NL	0	0	1	1	DE
aftenposten.no	News, media & entertainment	SE	5	0	48	5	SE, CA, US, DE, IE
nrk.no	News, media & entertainment	SE	10	0	7	1	SE
netflix.no	News, media & entertainment	GB	6	0	22	2	GB, CA
document.no	News, media & entertainment	DE	4	0	51	6	US, CA, SE, FR, DE, FI
klassekampen.no	News, media & entertainment	US	4	0	46	2	SE, US
spotify.no	News, media & entertainment	US	3	0	10	2	SE, US
worldofwarcraft.com	News, media & entertainment	IE	4	0	90	4	FR, SE, CA, US
dnt.no	Spare time / sports	GB	6	0	16	4	SE, CA, NL, FI
fotball.no	Spare time / sports	NL	8	0	113	5	SE, DK, CA, US, NL
skiforeningen.no	Spare time / sports	SE	1	0	14	3	NL, US, SE

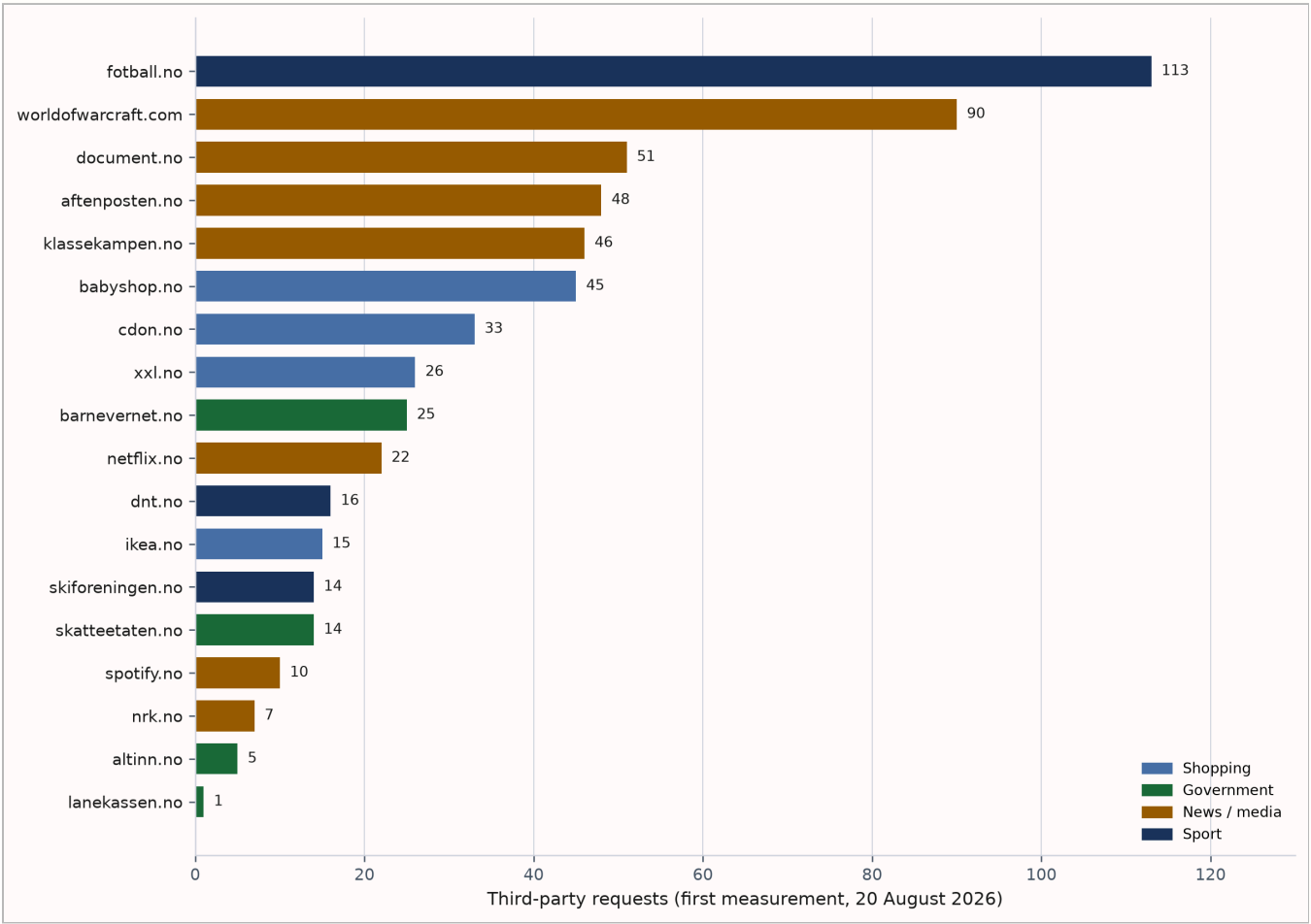
Note. Location of server is the KeyCDN country for the site's own IP. Third-party requests counts other hosts contacted during one homepage load. Countries excl. Norway counts unique third-party host countries after removing Norway; the list column may still include NO. Sector is added for comparison with Table 1 and Figure 4.

Table 3*Repeat Webb koll Measurement (20 August 2026) and KeyCDN Third-Party Countries (21 August 2026)*

Name of service	Sector	Location of server	Internal cookies	External cookies	3rd-party requests	Countries excl. Norway	List of countries
xxl.no	Shopping	US	3	0	26	2	IE, US
cdon.no	Shopping	US	4	0	33	1	US
ikea.no	Shopping	US	8	2	15	1	US
babyshop.no	Shopping	n/a	1	0	101	2	IE, US
skatteetaten.no	Government	US	5	0	14	4	FR, IE, SE, US
barnevernet.no	Government	NO	2	0	25	4	DE, IE, SE, US
altinn.no	Government	n/a	0	0	5	1	US
lanekassen.no	Government	NL	0	0	1	1	DE
aftenposten.no	News, media & entertainment	US	5	0	49	4	DE, IE, SE, US
nrk.no	News, media & entertainment	SE	11	0	7	1	SE
netflix.no	News, media & entertainment	US	6	0	22	1	US
document.no	News, media & entertainment	DE	4	0	51	5	FI, FR, IE, SE, US
klassekampen.no	News, media & entertainment	US	4	0	56	3	IE, SE, US
spotify.no	News, media & entertainment	US	3	0	10	2	SE, US
worldofwarcraft.com	News, media & entertainment	IE	4	0	92	3	IE, SE, US
dnt.no	Spare time / sports	n/a	6	0	16	5	FI, IE, NL, SE, US
fotball.no	Spare time / sports	NL	9	0	114	5	DK, IE, NL, SE, US
skiforeningen.no	Spare time / sports	US	1	0	14	3	IE, NL, US

Note. Repeat scan on the same day. Rankings in this report follow Table 2. n/a indicates that KeyCDN returned no country for the server IP. Measurements were taken at the landing host after any redirect.

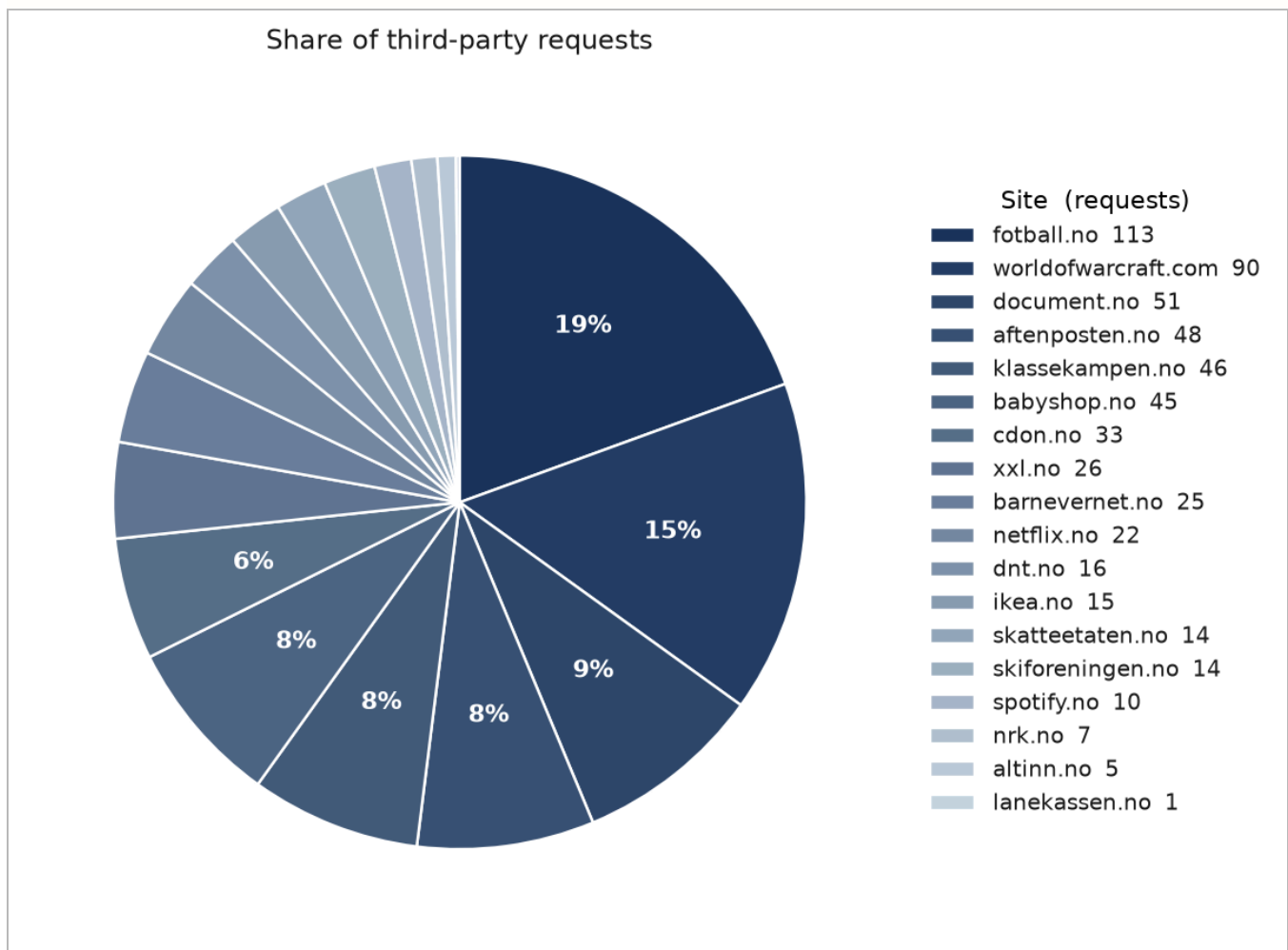
Figure 2
Third-Party Requests for 18 Sites (Table 2)



Note. Data from Table 2, ranked by third-party request volume.

Figure 3

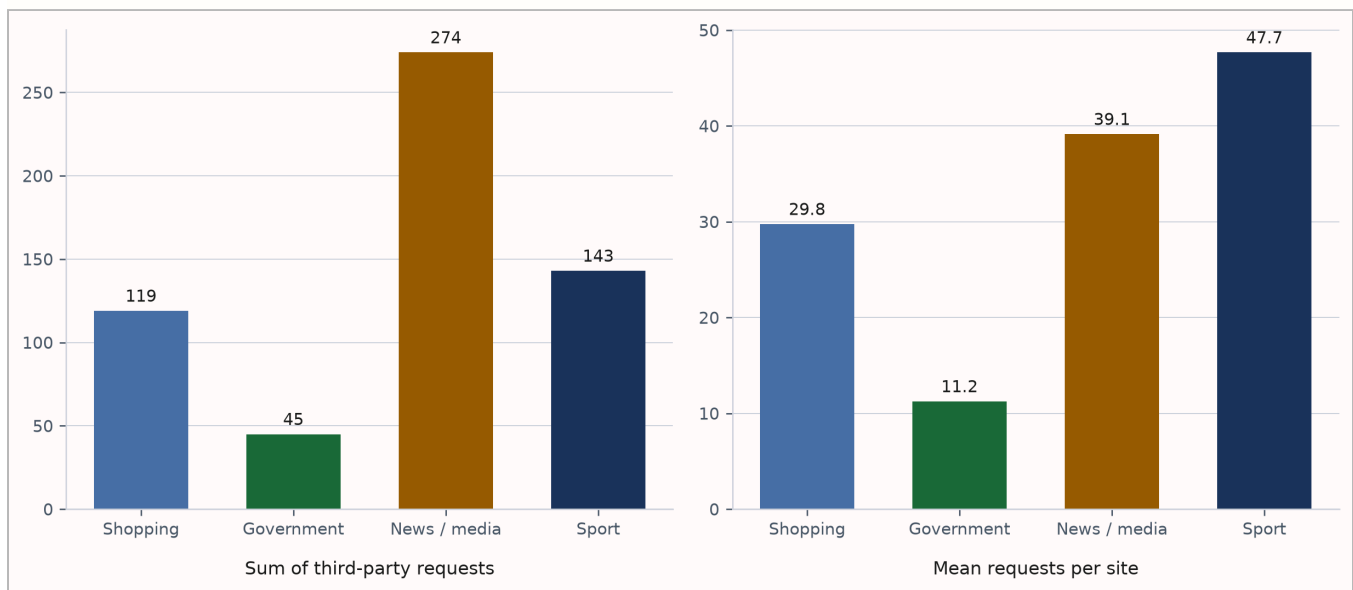
Share of Third-Party Requests (Table 2)



Note. Data from Table 2. fotball.no and worldofwarcraft.com together account for about one third of all recorded requests.

Figure 4

Third-Party Requests by Sector (Table 2)



Note. Data from Table 2, grouped by sector.

Rankings: highest contact, widest country spread, and restrained e-government

The assignment asks who shares most, who looks most global, and which e-government site is most restrained. Rank is Table 2 requests (Figure 2), not cookies, because 17 of 18 sites had zero external cookies. fotball.no is highest (113 requests, zero third-party cookies). document.no has the widest first country list (six excluding Norway). lanekassen.no (1) and altinn.no (5) are the quietest government fronts. Table 4 lists the top and bottom five. Table 3 would put babyshop.no at 101; we do not re-rank.

Table 4
Highest Five and Lowest Five by Third-Party Request Volume (Table 2)

Rank	Name of service	Sector	Req.	n countries
Highest 1	fotball.no	Spare time / sports	113	5
Highest 2	worldofwarcraft.com	News, media & entertainment	90	4
Highest 3	document.no	News, media & entertainment	51	6
Highest 4	aftenposten.no	News, media & entertainment	48	5
Highest 5	klassekampen.no	News, media & entertainment	46	2
Lowest 5	skiforeningen.no	Spare time / sports	14	3
Lowest 4	spotify.no	News, media & entertainment	10	2
Lowest 3	nrk.no	News, media & entertainment	7	1
Lowest 2	altinn.no	Government	5	1
Lowest 1	lanekassen.no	Government	1	1

Note. Ranked by third-party request volume in Table 2. skatteetaten.no (14 requests) is omitted because lower government sites appear in the lowest five. document.no has the widest country spread (6 excluding Norway). ikea.no is the only site with third-party cookies (2).

Case example: *klassekampen.no*

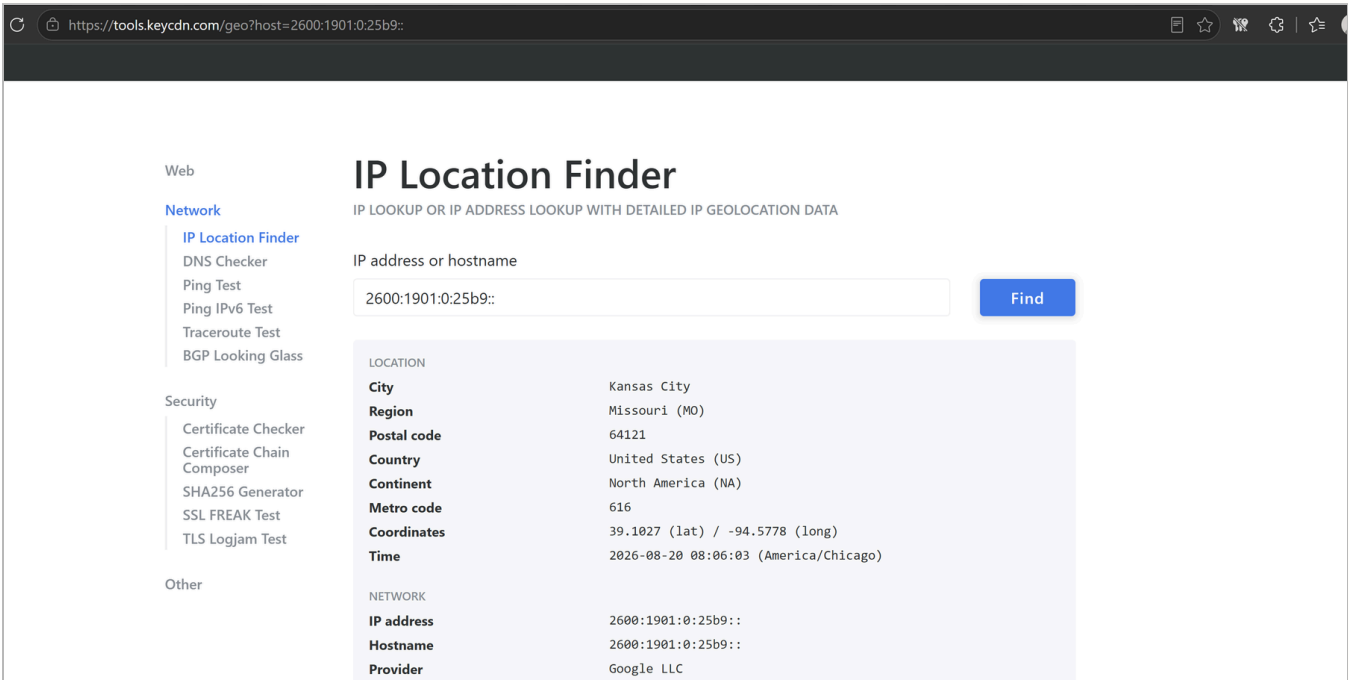
Figures 1, 5 and 6 show how a row is built. *klassekampen.no* is fifth in Table 4 (46 requests on Table 2). The screenshots are the later visit: four first-party cookies, zero third-party cookies, 56 requests to 13 hosts including Google ads, Meta and Cookiebot (Table 3).¹² The server IP maps to United States, Google (Figure 6).

Figure 5
*Webbkoll Summary for *klassekampen.no**



Note. Screenshot from Webbkoll (5th of July Foundation, n.d.-c; https://webbkoll.dataskydd.net/en). Scan time 20 August 2026, 12:36:46 UTC. Four first-party cookies, zero third-party cookies, 56 third-party requests to 13 hosts.

Figure 6
KeyCDN Lookup of the klassekampen.no Server IP



Note. Screenshot from KeyCDN IP Location Finder (KeyCDN, n.d.; <https://tools.keycdn.com/geo>). Lookup of 2600:1901:0:25b9:: returned United States (US), provider Google LLC. The same tool was applied to each unique third-party IP in the country list.

Part 2: Lawful basis for personal data processing

ICO lawful basis tool and scope

The ICO is the UK data protection authority. Its lawful-basis tool marks which Article 6 bases look APPROPRIATE for one activity (Information Commissioner's Office, 2026).¹⁴ We used the official Word report (26 August 2026). Name a purpose, then a basis (Information Commissioner's Office, n.d.). Tax data, optional cookies and checkout are different purposes. Mix them, and the tool mixes the answers. Each Table 5 row is one purpose from Table 1a.

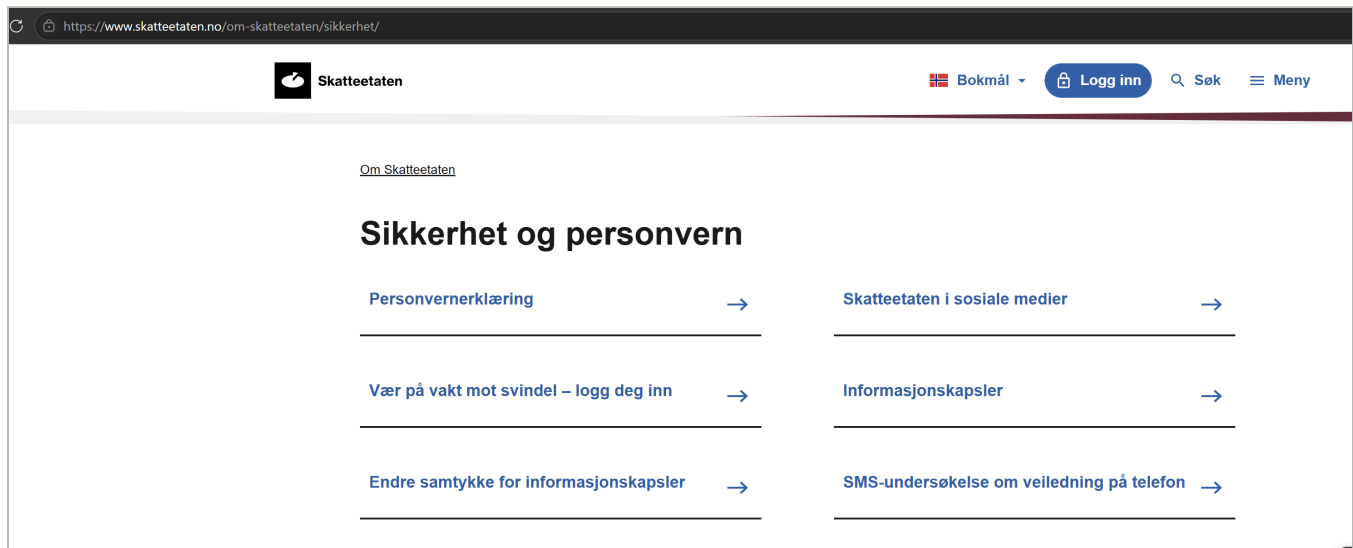
Skatteetaten.no: statutory tax data and optional cookies

The notice names two purposes. Mixing them in one ICO run mixes the bases. Identity, income and tax data are processed because the law requires it; opt-out is generally not possible (Norwegian Tax Administration, n.d.-a). ICO: legal obligation and public task APPROPRIATE. Consent NOT APPROPRIATE (likely invalid): there is no free, ongoing No to a statutory duty.

Optional statistics cookies (Skyra, Matomo, Google Analytics, Siteimprove) are the second purpose named in the notice. The notice claims consent. ICO: consent INCONCLUSIVE (controller, purpose and types not fully named). No basis APPROPRIATE. Table 5 has two rows. Figure 7 is the hub that already splits the privacy notice from the cookie page.

Figure 7

Security and Privacy Hub on skatteetaten.no



Note. Screenshot from the Norwegian Tax Administration security and privacy hub (Norwegian Tax Administration, n.d.-b; <https://www.skatteetaten.no/en/about-the-norwegian-tax-administration/security-and-privacy/>). The privacy notice and cookie page are separate destinations.

Netflix.no: paid streaming subscription

Purpose: account and payment data needed to provide the paid subscription (Netflix, 2026). ICO: contract APPROPRIATE. Consent likely invalid: no real choice if the service cannot run without the account. Ads and marketing in the same notice are other purposes and were left out.

Fotball.no: match history publication

Norges Fotballforbund publishes name and club of active players aged 13+, with club opt-out (Norges Fotballforbund, n.d.). ICO: legitimate interests APPROPRIATE. NFF is not a public authority, so public task does not fit. FIKS membership may rest on contract.¹⁵ That is another purpose and was not this ICO run.

Document.no: Google Signals advertising analytics

Google Signals is advertising analytics.¹⁶ Article 6 is not named (Document Media AS, n.d.). Collection is said to run only with a Google login and ad personalization. That looks like consent. ICO: no basis APPROPRIATE. Consent INCONCLUSIVE: the request is not clear, prominent and separate; the choice sits in Google settings; Document Pluss terms also bundle the notice. Contract and legitimate interests do not fit this advertising purpose. Table 5: Partial.

Babyshop.no: checkout and order fulfilment

Purpose: name, address, contact, order and payment to deliver goods. The sales terms are a purchase contract (Babyshop Sthlm Holding AB, n.d.). ICO: contract APPROPRIATE. The notice does not label Article 6(1)(b). Marketing stays a separate purpose.

Table 5: match criteria and partial results

Each row is one purpose. Yes: notice and ICO mark line up. Partial: the notice aims at consent; ICO marks consent INCONCLUSIVE and no basis APPROPRIATE.

Table 5*ICO Lawful Basis Summary for Five Selected Services*

Service	Processing purpose (one)	Basis claimed in notice	ICO result	Match?
skatteetaten.no	Tax and registry administration (identity, income, tax data)	Legal obligation / processing required by law; opt-out generally not possible	Legal obligation and public task APPROPRIATE; consent NOT APPROPRIATE (likely invalid)	Yes
skatteetaten.no	Optional web statistics cookies	Consent	Consent INCONCLUSIVE; no basis APPROPRIATE	Partial (separate run)
netflix.no	Account and payment data to provide paid streaming	EEA/UK: contractual necessity to provide the service to members	Contract APPROPRIATE; other bases NOT APPROPRIATE	Yes
fotball.no	Publishing player name and club for match history (13+)	Public-interest match history with club opt-out (LI in practice)	Legitimate interests APPROPRIATE; other bases NOT APPROPRIATE	Yes for this purpose
document.no	Google Signals / advertising analytics from site activity	Article 6 not named; Google login plus ad personalization; opt-out links	Consent INCONCLUSIVE; no basis APPROPRIATE	Partial
babyshop.no	Customer, delivery and payment data to complete a purchase	Not labeled as Article 6(1)(b); purchase data processed to fulfil orders	Contract APPROPRIATE; other bases NOT APPROPRIATE	Yes for checkout

Note. ICO Word reports dated 26 August 2026 (Information Commissioner's Office, 2026). Each row covers one processing purpose only.

Conclusion

Part 1: fotball.no has the most third-party requests (113) and zero third-party cookies. document.no maps to the most countries. lanekassen.no and altinn.no are the quietest e-government fronts. News/media is the highest sector; government the lowest. A request count is not a cookie count.

Part 2: four core purposes match the notice (tax, Netflix, match lists, checkout). Two consent tests fail: skatteetaten optional cookies and document.no Google Signals. One purpose needs one basis. Together the two parts show how the page loads, and why the controller may use the data.

Footnotes

¹ GDPR is the General Data Protection Regulation, the main EU law on personal data. It also applies in Norway through the EEA. Article 6 lists the allowed reasons for using personal data (legal obligation, public task, contract, consent, legitimate interests, vital interests). [^]

² Webbkoll is a free web site checker run by the Swedish 5th of July Foundation. You type a URL. It opens that page once and reports cookies, other hosts contacted, HTTPS and some security headers. It does not log in. [^]

- ³ ICO is the Information Commissioner's Office, the UK data protection authority. The "lawful basis interactive guidance tool" is a questionnaire on the ICO web site. After the questions it produces a Word report that marks each Article 6 basis APPROPRIATE, NOT APPROPRIATE or INCONCLUSIVE. We used that Word file as the official result. ^
- ⁵ A host is a machine with a name on the internet, such as fonts.googleapis.com. A third-party request means the open page asked that other machine for something (a script, an image, an ad). Counting requests is not the same as counting cookies, and it is not the same as proving that a name or a tax file was sent. ^
- ⁶ Chromium is the open-source browser engine behind Google Chrome. Webbkoll uses a clean Chromium session so the scan is close to a first visit with no extra extensions. ^
- ⁷ HTTPS is the encrypted version of web traffic (the padlock in the browser). HTTP without the S is the unencrypted version. Webbkoll checks whether the site forces HTTPS. ^
- ⁸ An IP address is a number that identifies a machine on the internet, for example 192.0.2.1 or an IPv6 string such as 2600:1901:0:25b9:: Webbkoll prints the IP of the site it reached. We then paste that number into KeyCDN (see footnote 10). ^
- ⁹ Do Not Track (DNT) is an old browser setting that asked sites not to track the user. Many sites ignore it. We left it off so the scan matches a default browser, as Webbkoll's FAQ describes. ^
- ¹⁰ KeyCDN is a commercial content-delivery company. It also offers a free "IP Location Finder" at tools.keycdn.com/geo. You paste an IP address. The page returns a likely country, city and provider (for example Google LLC, United States). That answer is a geolocation guess. It can be wrong when the IP belongs to a cloud, a VPN or an anycast network. We used it because the assignment asks for server location and for countries of third-party hosts. It is not a government register of data exports. ^
- ¹² Cookiebot appeared as one of the third-party hosts on the klassekampen.no Webbkoll result. We record it as a contacted host, not as a separate investigation. ^
- ¹⁴ In the ICO Word report, APPROPRIATE means that basis can work for the purpose you described. NOT APPROPRIATE means it should not be used. INCONCLUSIVE means the facts you entered were not enough for a Yes (often the consent request is unclear). Those labels are guidance, not a court judgment. ^
- ¹⁵ FIKS is Norges Fotballforbund's club and player administration system. A membership or licence in FIKS can rest on a contract with the club. Publishing a name on a public match list is a different activity, so it can have a different Article 6 basis. ^
- ¹⁶ Google Signals is a Google Analytics feature. When a visitor is logged into Google with ads personalization on, Google can join activity across devices and add demographic and interest reports. That is advertising analytics, not the same as a newspaper subscription contract. ^

References

Babyshop Sthlm Holding AB. (n.d.). *Integrity policy*. Babyshop. Retrieved August 26, 2026, from <https://www.babyshop.com/en-us/information/integrity-policy>

Document Media AS. (n.d.). *Personvernerklæring*. Document.no. Retrieved August 26, 2026, from <https://www.document.no/personvernerklaering/>

5th of July Foundation. (n.d.-a). *About*. Webbkoll. Retrieved August 20, 2026, from <https://webbkoll.dataskydd.net/en/about>

5th of July Foundation. (n.d.-b). *FAQ*. Webbkoll. Retrieved August 20, 2026, from <https://webbkoll.dataskydd.net/en/faq>

5th of July Foundation. (n.d.-c). *Webbkoll* [Computer software]. Retrieved August 20, 2026, from <https://webbkoll.dataskydd.net/en>

General Data Protection Regulation. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation). EUR-Lex. <https://eur-lex.europa.eu/eli/reg/2016/679/oj>

Information Commissioner's Office. (2026). *Lawful basis interactive guidance tool*. <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/lawful-basis/lawful-basis-interactive-guidance-tool/>

Information Commissioner's Office. (n.d.). *A guide to lawful basis*. Retrieved August 26, 2026, from <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/lawful-basis/a-guide-to-lawful-basis/>

KeyCDN. (n.d.). *IP Location Finder* [Computer software]. Retrieved August 21, 2026, from <https://tools.keycdn.com/geo>

Netflix. (2026, April 10). *Privacy statement*. <https://www.netflix.com/privacy>

Norges Fotballforbund. (n.d.). *Personvern*. Retrieved August 26, 2026, from <https://www.fotball.no/personvern>

Norwegian Tax Administration. (n.d.-a). *Privacy policy*. Retrieved August 26, 2026, from <https://www.skatteetaten.no/en/about-the-norwegian-tax-administration/security-and-privacy/privacy-policy/>

Norwegian Tax Administration. (n.d.-b). *Security and privacy*. Retrieved August 26, 2026, from <https://www.skatteetaten.no/en/about-the-norwegian-tax-administration/security-and-privacy/>